

COMMUNICATIONS 2025-2026's 1ST SEMSTER MID EXAM:

Question No. 1 (24 Marks):

Q1. Mesh topology with 4 stations, how many connections in this topology?

- A) 6
- B) 12
- C) 3

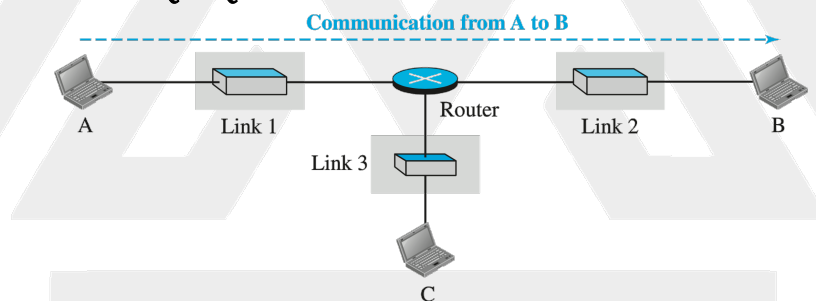
Q2. The TCP/IP suite is:

- A) unidirectional communication system
- B) bidirectional communication system
- C) multidirectional communication system

Q3. The TCP/IP suite is:

- A) software layers
- B) software and hardware layers
- C) hardware layers

For the following figure "Answer Q.4-Q.5":



Q4. The router is involved in:

- A) all five layers
- B) only three layers
- C) only two layers

Q5. The station that makes no encapsulation/decapsulation is:

- A) sender
- B) router
- C) link

Q6. The addresses of the source & destination hosts are added as a header to the message.

- A) network layer
- B) data-link layer
- C) transport layer

Q7. Changing TV channels by IR remote control is:

- A) point-to-point half-duplex communication
- B) point-to-point simplex communication
- C) point-to-point full-duplex communication

Q8. If the network stores and forwards a packet, then this network is called:

- A) switched network
- B) circuit-switched network
- C) packet-switched network

Q9. DSL service is an upgraded service with respect to:

- A) Dial-up service
- B) Wi-Fi
- C) ISP

Q10. Variation in the packet arrival time is called:

- A) timeliness
- B) jitter
- C) timing

Q11. The bandwidth of a TV channel is usually measured in:

- A) KHz
- B) GHz
- C) MHz

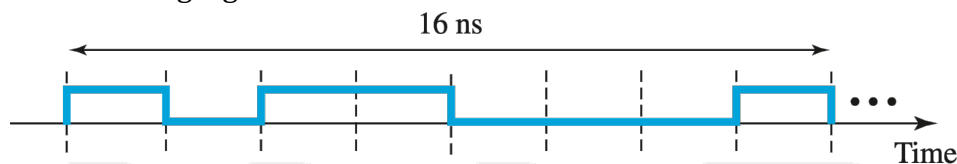
Q12. The channel in broadband transmission is called:

- A) low-pass channel
- B) band-pass channel
- C) pass-band channel

Q13. The spectrum of voice signals is:

- A) discrete
- B) continuous
- C) non-periodic

Q14. The bit rate of the following signal is:



- A) 500 mbps
- B) 500 μ bps
- C) 500 Mbps

Q15. We measure the performance of a telephone line (4 kHz of bandwidth). When the signal is 10 V, the noise is 5 mV. What is the maximum data rate supported by this telephone line?

- A) 43866 bps
- B) 43863 bps
- C) 43.866 bps

Q16. What is the phase shift of a sin wave with maximum amplitude after 1/4 cycle?

- A) 90 degrees
- B) 0 degrees
- C) -90 degrees

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Ans	B	B	A	B	C	A	B	C	A	B	C	B	B	C	B	C

Question No. 2 (6 Marks):

What is the length of a bit in a channel with a propagation speed of 2×10^8 m/s if the channel bandwidth:

- A) 1 Mbps
- B) 10 Mbps
- C) 100 Mbps

Solution:

$$\text{Bit Length} = \text{Propagation Speed} \cdot \text{Time}$$

A)

$$\text{Bit Length} = 2 \times 10^8 \text{ m/s} \cdot \frac{1}{1 \times 10^6 \text{ bps}} = 200 \text{ m for each bit}$$

B) 10 Mbps

$$\text{Bit Length} = 2 \times 10^8 \text{ m/s} \cdot \frac{1}{10 \times 10^6 \text{ bps}} = 20 \text{ m for each bit}$$

C) 100 Mbps

$$\text{Bit Length} = 2 \times 10^8 \text{ m/s} \cdot \frac{1}{100 \times 10^6 \text{ bps}} = 2 \text{ m for each bit}$$

References Pictures

Al-Balqa Applied University

جامعة البلقاء التطبيقية

Q1: (Please write your answers into table only, otherwise your answers will be neglected) (24 marks)

Question	1	2	3	4	5	6	7	8	9	10
Answer	a	b	a	b	a	b	c	a	a	a
Question	11	12	13	14	15	16				
Answer	c	a	b	c	a	a				

+12

1. Mesh topology with 4 stations, how many connections in this topology:

a) 6 b) 12 c) 3

2. The TCP/IP suite is: 5 layers

a) unidirectional communication system
b) bidirectional communication system
c) multidirectional communication system

3. The TCP/IP suite is:

a) software layers
b) software and hardware layers
c) hardware layers

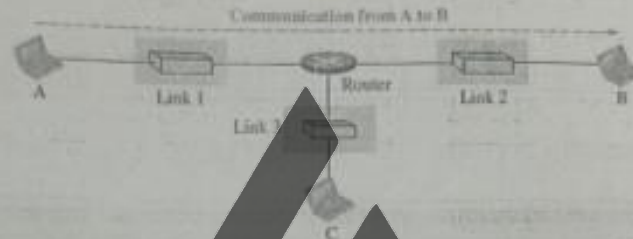
4. The router in the following figure is involved in:

a) all five layers b) only three layers c) only two layers

5. Due to the following figure, the station that makes no encapsulation/decapsulation is:

a) sender b) router c) link

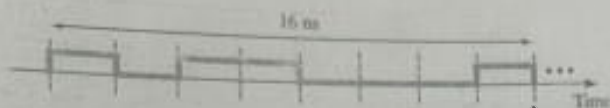
Both



6. The addresses of the source & destination hosts are added as a header to the message ^{layer 1} Application layer _{ir.}
- a) network layer ^{layer 3} b) data-link layer ^{layer 4} c) transport layer ^{layer 2}
7. Changing TV channels by IR remote control is Point-to-point simplex
- a) point-to-point half-duplex communication
 b) point-to-point simplex communication
 c) point-to-point full-duplex communication
8. If the network stores and forwards a packet, then this network is called:
- a) switched network
 b) circuit-switched network
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9. DSL service is an upgraded service with respect to:
- a) dial-up service b) Wi-Fi c) ISP
10. Variation in the packet arrival time is called:
- a) timeliness b) jitter c) timing
11. The bandwidth of a TV channel is usually measured in:
- a) kHz b) GHz c) MHz
12. The channel in broadband transmission is called:
- a) low-pass channel
 b) band-pass channel
 c) pass-band channel



- ✓✓ 13. The spectrum of voice signals is:
 a) discrete (b) continuous c) non-periodic
- ✓✓ 14. The bit rate of the following signal is:



$$\frac{8}{16 \text{ ns}} = 500 \text{ M}$$

- a) 500 mbps b) 500 μ bps (c) 500 Mbps

- ✓✓ 15. We measure the performance of a telephone line (4 KHz of bandwidth). When the signal is 10 V; the noise is 5 mV. What is the maximum data rate supported by this telephone line?

$$\text{SNR} = 20$$

$$\text{SNR}_{dB} = 33$$

$$C = B \cdot \log_2(1 + \text{SNR})$$

$$= 4 \text{ kHz} \cdot \log_2(1 + 20)$$

$$= 43866$$

- (a) 43866 bps b) 43863 bps c) 43.866 bps

- ✓✓ 16. What is the phase shift of a sin wave with maximum amplitude after 1/4 cycle:

- (a) 90 degrees b) 0 degrees c) -90 degrees

Q2: What is the length of a bit in a channel with a propagation speed of 2×10^8 m/s if the channel bandwidth is

- a. 1 Mbps?
 b. 10 Mbps?
 c. 100 Mbps?

$$T = \frac{1}{F}$$

(6 marks)

$$\text{Bit Length} = \frac{\text{Prop. speed}}{\text{Bandwidth}}$$

$$a) \text{ Bit Length} = 2 \times 10^8 \times \frac{1}{1 \text{ M}} \Rightarrow 2 \times 10^8 \times 1 \times 10^{-6} = 200 \text{ m}$$

$$b) \text{ Bit Length} = 2 \times 10^8 \times \frac{1}{10 \text{ M}} \Rightarrow 2 \times 10^8 \times 10 \times 10^{-6} = 20 \text{ m}$$

$$c) \text{ Bit Length} = 2 \times 10^8 \times \frac{1}{100 \text{ M}} \Rightarrow 2 \times 10^8 \times 100 \times 10^{-6} = 2 \text{ m}$$

GOOD LUCK